Edward Girling

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SUMMARY

Senior Software Engineer and Data Scientist with experience in fintech, defense, and open-source (Click 17.5k Stars) . Expert in Python and AWS ECS. Proven track record of owning products and teams from feature design to production.

PROFESSIONAL EXPERIENCE

Wallstreet.io | **Senior Software Engineer** | *06/2022 – Present*
Lead full stack developer who drives product initiatives from concept to deployment

- Modernized platform documentation using Gen AI workflows. Workflows also update documentation as software changes are made. Relevant information is surfaced through a MCP server written in Python.
- Owned a mission-critical stock strategy performance platform with 99.99% uptime and implemented a comprehensive automated testing suite to improve deployment confidence and system stability.
- Designed and launched core trading platform features including a real-time stock price alert system, stock opportunity screener, and stock strategy explorer, directly supporting product growth and user engagement.
- Built a high-throughput alerting platform using Python, PHP, AWS, and event-driven infrastructure to monitor millions of market trades and deliver low-latency notifications.
- Optimized AWS infrastructure utilization and application performance, reducing compute costs by 30% while improving application response times and page-load performance. Dockerized all infrastructure and implemented dynamic scaling rules which reduced computer cost by an additional 35 percent.
- Reduced critical production errors from thousands per day to fewer than 10 through targeted instrumentation, root-cause analysis, and large-scale bug remediation efforts.
- Led platform reliability initiatives by implementing end-to-end uptime monitoring, infrastructure health checks, and critical service instrumentation, significantly improving operational visibility and incident response.
- Directed modernization of legacy infrastructure across Redis, Elasticsearch, MySQL, PostgresSQL and Laravel while maintaining full feature parity and improving platform security, scalability, and maintainability.

Teledyne Brown Engineering (Navy Bureau of Medicine) | **Data Scientist** | *12/2020 – 05/2022*

- Delivered analytics infrastructure and reporting capabilities that directly supported renewal of an $8M government contract by demonstrating measurable operational value.
- Developed Python-based data pipelines powering Navy Readiness and Manpower dashboards, reducing metric generation times by up to 93% and lowering monthly operational costs by thousands of dollars.
- Performed large-scale sentiment analysis on sailor job satisfaction surveys using Scikit-Learn and Pandas generating executive-level visualizations and insights presented to Navy leadership.

Innovative Computer Solutions (ICS) | **Software Engineer** | *01/2020 – 11/2020*

- Developed and deployed generative text APIs using PyTorch and custom Machine Learning pipelines on AWS.
- Reduced API latency by 50% through targeted cloud hardware optimization and model profiling.

United States Navy | **Division Officer** | *05/2014 – 12/2019*

- Designed and developed a security-hardened, low-bandwidth operational web application to improve work-sharing in constrained environments; presented the proof-of-concept directly to senior leadership.

EDUCATION & OTHER

- **Open Source:** Lead maintainer of **Click** (Python CLI package), 17,100+ GitHub Stars, https://github.com/pallets/click/ . Responsible for triage, feature development, and community management.
- **Writing:** 10,000 words on Transformer Architecture (ML). 40,000 words on designing command line apps in Python. 2,500 words on improving technical documentation quality.
- **United States Naval Academy** | **B.S. in Applied Mathematics** | *2014*